

Project Report: Multi-Object Tracking and Behavioral Analysis of Dogs

1. Introduction

The field of computer vision has made significant strides in Multi-Object Tracking (MOT), moving beyond simple object detection to understanding the behavior and predicting the movement of multiple agents within a dynamic environment. This capability has profound applications in areas such as autonomous surveillance, human-robot interaction, and behavioral analysis in wildlife.

This project addresses the specific challenge of tracking multiple dogs and people in real-world video footage. The primary objective is to develop a robust system that can:

1. Reliably detect and differentiate between dogs and people in complex urban scenes.
2. Assign and maintain a unique identity for each detected dog across consecutive video frames, even during brief occlusions.
3. Analyze the movement patterns of each tracked dog to classify its behavior into distinct states: **stationary**, **moving toward a user**, or **moving elsewhere**.
4. Predict and visualize the short-term future trajectory of each dog.

To achieve this, the project implements a state-of-the-art "Tracking-by-Detection" pipeline, leveraging a custom-trained **YOLOv8** model for object detection and the **DeepSORT** algorithm for robust tracking and re-identification.

2. Literature Survey

The Tracking-by-Detection paradigm is the dominant approach in modern multi-object tracking. This methodology decouples the problem into two main stages: detection and association.

- **Object Detection:** Early methods relied on two-stage detectors like the R-CNN family, which were accurate but slow. The advent of single-stage detectors, most notably the **YOLO (You Only Look Once)** family, revolutionized real-time object detection. YOLO treats detection as a regression problem, directly predicting bounding boxes and class probabilities from a full image in a single pass. This project utilizes **YOLOv8**, the latest iteration, known for its exceptional balance of speed and accuracy, making it ideal for video processing. To achieve high accuracy for a specific domain, transfer learning is employed by fine-tuning a pre-trained model on a custom dataset.
- **Data Association and Tracking:** Once objects are detected in each frame, they must be associated with existing tracks.

- **SORT (Simple Online and Realtime Tracking):** This foundational algorithm introduced the use of a **Kalman Filter** to predict an object's position in the next frame based on its current motion model. The **Hungarian algorithm** is then used to solve matching predicted bounding boxes with newly detected boxes based on Intersection over Union (IoU) distance. However, SORT is prone to identity switches, especially after long occlusions.
- **DeepSORT:** This algorithm extends SORT by integrating an **appearance model**. In addition to motion prediction, a deep Convolutional Neural Network (CNN) is used to extract a feature vector (or "embedding") from each detected object. The cost matrix for the Hungarian algorithm is then augmented with a cosine distance metric between appearance embeddings. This allows DeepSORT to re-identify an object even if it reappears far from its last known location, making it much more robust against occlusions and identity switches. This project leverages the `deep-sort-realtime` library, which provides an efficient implementation of this methodology.

3. Methodology

The project was implemented as a Python application within a Google Colab environment, following a clear, multi-stage pipeline.

3.1 Data Acquisition and Model Training

- **Dataset:** A custom dataset, "Stray Dogs," was sourced from **Roboflow**. This dataset contained thousands of images with labeled bounding boxes for various classes of dogs and people.
- **Model Selection:** The **YOLOv8s** (small) model was chosen as the base for its strong balance of accuracy and performance.
- **Training:** The model was trained for **50 epochs** on the custom dataset using the `ultralytics` framework. This fine-tuning process adapted the model's weights to become highly proficient at detecting dogs and people in varied environments, significantly outperforming generic pre-trained models.

3.2 The Tracking Pipeline (Per-Frame)

For each frame of the input video, the following steps are executed:

1. **Detection:** The custom-trained YOLOv8s model processes the frame and generates a list of potential detections, each with a bounding box, a class ID, and a confidence score. Detections below a confidence threshold of `0.40` are discarded to reduce noise.

2. **Class ID Mapping:** The multiple, inconsistent "dog" labels from the training data (e.g., 'dog', 'Dog', '0') are dynamically mapped to a single internal class ID (0 for dog, 1 for person) for clean and consistent downstream logic.
3. **Tracking with DeepSORT:** The filtered detections are passed to the `DeepSort` tracker instance. The tracker performs the following:
 - **Prediction:** The Kalman Filter for each existing track predicts its new position.
 - **Association:** A cost matrix is built using both motion (from Kalman) and appearance data. The Hungarian algorithm solves this matrix to match detections to the most likely tracks.
 - **Lifecycle Management:** Tracks that are consistently matched with detections are confirmed. Unmatched detections are used to initialize new tracks (after `n_init=5` consecutive hits). Confirmed tracks that are lost for `max_age=50` frames are deleted.
4. **Behavioral & Trajectory Analysis:**
 - **History Tracking:** The centroid coordinates and bounding box dimensions for each confirmed track are stored in a history queue.
 - **Behavior Classification:** For each dog, a rule-based function analyzes its recent history to determine its state (`stationary`, `moving_toward_user`, `interacting`).
 - **Trajectory Prediction:** The average velocity over the last 5 frames is calculated to predict and generate a smooth future path of 15 points.
5. **Visualization and Output:**
 - A copy of the original frame is used to draw all visual elements.
 - **Bounding Boxes:** Green for dogs, blue for people.
 - **Labels:** Each box is annotated with its Track ID, behavioral state, and confidence score.
 - **Past Trails:** A light blue polyline is drawn connecting the last 20 historical centroids of each dog.
 - **Future Trajectory:** A red polyline is drawn showing the predicted future path.
 - **Dashboard:** A statistics overlay shows real-time counts of tracked dogs and their states.
 - The final annotated frame is written to an output MP4 video file.
6. **Logging:** All tracking data for each frame is saved as a JSON object in a `.jsonl` log file for further analysis.

4. Results

The system was tested on a real-world video clip featuring two dogs and multiple people in a narrow city street (*1088787659-preview.mp4*). The results successfully demonstrate the achievement of all project objectives.

- **Detection and Tracking Performance:** The custom-trained YOLOv8s model successfully detected and tracked both dogs and all relevant people in the frame. The DeepSORT tracker, with its tuned `max_age` and `n_init` parameters, proved highly effective at maintaining consistent track IDs without flickering, even during partial occlusions.
- **Behavioral Classification:** The rule-based system correctly identified the behavioral states.
 - When the dogs were standing still at the beginning and end of the clip, they were correctly labeled as `stationary`.
 - When they began to jump and play, their state correctly switched to `moving_elsewhere`.
 - The `interacting` state was also successfully triggered when their bounding boxes came into close proximity.
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- **Visual Output:** The final video output clearly displays all the required information. The bounding boxes are stable, the IDs are consistent, the behavioral labels are accurate, and the past/future trajectories provide excellent visual context for the dogs' movement. The dashboard offers an at-a-glance summary.

(Please refer to the "Results Folder" section below for generated output (output_video_final_stable_1.mp4)).

5. Conclusion and Future Work

This project successfully developed and implemented a multi-object tracking and behavioral analysis system for dogs. By combining a custom-trained YOLOv8 detector with the robust DeepSORT algorithm, the system achieves accurate detection and stable tracking. The rule-based behavioral analysis provides valuable, high-level insights into the dogs' actions, fulfilling the core requirements of the problem statement.

Future Work:

- **Advanced Behaviors:** Incorporate pose estimation (OpenPifPaf) for complex behaviors.
- **Real-Time Deployment:** Optimize for edge devices like NVIDIA Jetson Nano.
- **Robustness:** Improve re-identification for similar-looking dogs.